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Code to Connect
Hackathon 2026

AI RISK ENGINE

From Financial News to Portfolio Decisions

Turning unstructured financial information into explainable, systematic portfolio risk signals.

Candidate	**Mridul Rajgaria**
Institution	Vellore Institute of Technology, Bhopal
Event	Code to Connect Hackathon 2026 • Final Round
Implemented application	Module A — Tactical Index Rebalancer
Portfolio universe	14 large-cap equities

NEWS HEADLINE • ILLUSTRATIVE EXAMPLE

Acme Corp faces regulatory probe over accounting practices

↓

STRUCTURED RISK SIGNAL

event_type	**Regulatory**
sentiment_score	-0.80
impact_score	**7.85 / 10**
trading_action	REDUCE

Illustrative values, not an application output

Text in, structured signal out, portfolio weight adjusted.

Financial information arrives as text. A portfolio needs numbers.

RAW INFORMATION • ILLUSTRATIVE EXAMPLE

News headline

Acme Corp faces regulatory probe over accounting practices

Social post

$ACME this probe is going to get ugly. Not touching it.

Market data

ACME daily prices (yfinance)

STRUCTURED RISK SIGNAL • ILLUSTRATIVE VALUES

`event_type`	**Regulatory**	1 of 8 classes, rule-based
`sentiment_score`	`-0.80`	FinBERT tone, range [-1.0, +1.0]
`confidence`	`0.90`	softmax conviction, [0.0, 1.0]
`impact_score`	**7.85**	severity, [1.0, 10.0]: Event 35% + Sentiment 30% + Credibility 20% + Volume 15%
`trading_action`	REDUCE	downstream suggestion: BUY, REDUCE or HOLD

The engine converts unstructured financial text into consistent, explainable information that a portfolio strategy can consume.

Information overload

Too many headlines and posts to assess consistently by hand.

Tweets, NewsAPI and GDELT are processed into one schema with cross-label deduplication.

Black-box distrust

A score is less useful if its reasoning cannot be inspected.

A linear impact score (1.0 to 10.0) built from 4 named, audit-ready components, configurable via YAML.

Execution drag and noise

Reacting to every small change can cause excessive trading.

Rebalancer with ±0.20 deadband, EMA half-life 2.0 days and a 10% daily turnover cap.

Reproducibility

A reviewer needs to reproduce the result without live APIs.

Deterministic committed sample datasets (<5 MB): full audit and replay with zero API keys or live downloads.

One pipeline: collect, understand, decide, deliver

1 Collect

Gather headlines, social posts and market data.

Kaggle Stock Tweets (2021–22), NewsAPI, GDELT 2.0, yfinance • 14 core + 7 news-only equities

2 Understand

Identify the company, read sentiment, classify the event.

Normalization, cashtag linking, FinBERT (CPU), event classifier • 1–10 impact severity

3 Decide

Turn the signal into a bounded weight adjustment.

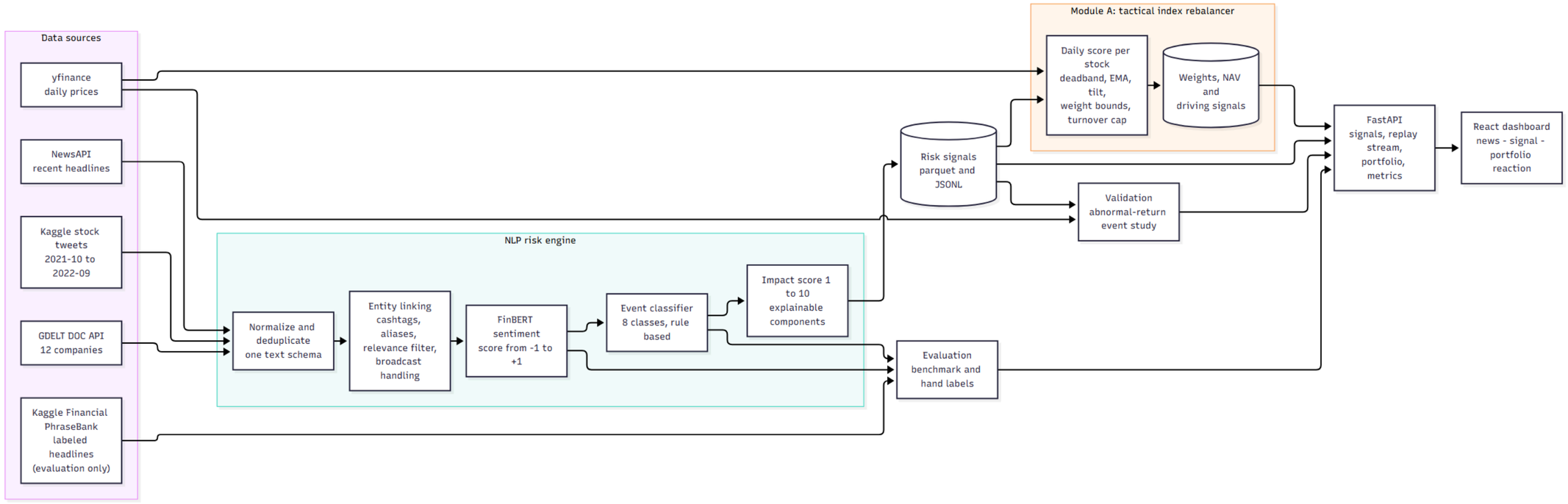
EMA smoothing, tilt k=1.5, negative multiplier 1.25x • weights [3.57%, 14.29%], 10% turnover cap

4 Deliver

Show the result and its reasoning in a dashboard.

FastAPI, Server-Sent Events, React 18, Apache ECharts • News Stream, Signal Inspector, Portfolio Reaction

Technical architecture, as implemented (unchanged from the original submission)



How one headline becomes a risk signal

Illustrative example: values follow the published formula but are not an application output.

1 • Headline

Acme Corp faces regulatory probe over accounting practices

First cleaned: template spam stripped, URLs normalised, HTML entities decoded.

2 • Entity linking

Acme Corp → $ACME

The system determines which company the text refers to.

cashtags, aliases, name dictionaries, broadcast splitting

3 • Sentiment

`-0.80`

`-1.0` `+1.0`

The model estimates whether the language is positive or negative. Confidence 0.90.

FinBERT (ProsusAI/finbert), SQLite cache

4 • Event type

Regulatory

Classifies the event: earnings, regulatory action, credit event, etc.

8 rule-based classes • precision 1.0 on Credit / Regulatory / Earnings triggers (180 hand-labelled texts; recall 1.00 / 0.68 / 0.23)

5 • Impact scoring

Event severity	`35% Regulatory = 0.9`
Sentiment strength	`30% |s| × c = 0.72`
Source credibility	`20% GDELT 0.85`
Mention volume	`15% 0.40 (illustrative)`

Credibility: GDELT 0.85, NewsAPI 0.80, Kaggle Tweets 0.50. Severity: Regulatory/Credit 0.9, Earnings 0.7, Product 0.5, Other 0.3. Volume: log-scaled tweet density vs rolling median.

IMPACT-SCORE FORMULA

$$\text{Impact} = 1.0 + 9.0 \times \sum(w_i \times \text{Component}_i) \in [1.0, 10.0]$$

$$1.0 + 9.0 \times (0.35{\times}0.90 + 0.30{\times}0.72 + 0.20{\times}0.85 + 0.15{\times}0.40) = 1.0 + 9.0 \times 0.761 = \mathbf{7.85}$$

Combines event importance, sentiment, source reliability and mention activity into one consistent severity measure.

What the score means, and does not

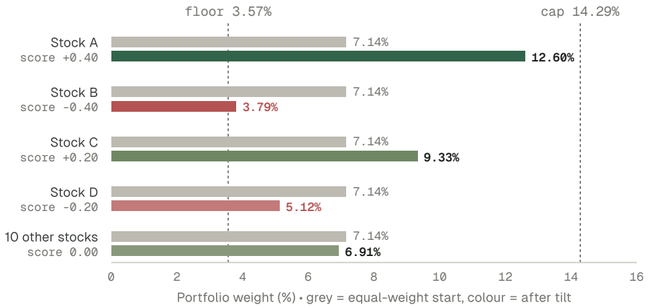
It ranks how severe a piece of text looks. It is not a probability of a price move or a promise of trading return.

Every score on the dashboard can be unpacked via “Show the math” into its 4 raw component values.

How a risk score nudges portfolio weights, within limits

If all 14 stocks start with equal weights, each is 1 / 14 ≈ 7.14% of the portfolio. Signals tilt that baseline; rules keep the tilt bounded.

Equal-weight portfolio → Risk signal → Bounded adjustment



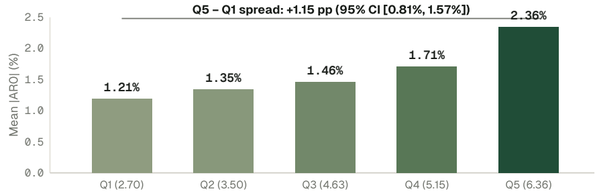
Conceptual example, not an application output. 14 hypothetical stocks with invented smoothed scores (A +0.40, B −0.40, C +0.20, D −0.20, ten at 0), run through the exponential tilt with k = 1.50. Here neither the weight bounds nor the 10% turnover cap binds (one-way turnover 7.65%); they would limit more extreme scores. Not an optimal-allocation claim.

1 • Daily score	Per signal: |s| < 0.20 is ignored (deadband); negative s is multiplied by 1.25. Signals are averaged, weighted by attribution × impact / 10.
2 • EMA smoothing half-life 2.0 days	score = (1−α)·prev + α·raw, α = 1 − 2^(−1/2) ≈ 0.293. A brief burst of news cannot cause an immediate swing.
3 • Exponential tilt k = 1.50	w ∝ exp(k × score), normalised to sum to 1. Positive scores raise a stock above 7.14%; negative scores lower it.
4 • Weight bounds 3.57% to 14.29%	Weights are projected into 0.5× to 2.0× of the 7.14% baseline, with the excess redistributed, so no stock gets too small or too large.
5 • Turnover cap 10% one-way / day	If the move exceeds the cap, trades are scaled down; weights are then renormalised to exactly 1.0.
Costs, fallback 5 bps	Backtest deducts 5 bps on turnover from net NAV. Missing market data reverts to equal weight, 7.14%.

System performance: price move association vs. strategy returns

Evaluation across 2,072 ticker-days and 252-day backtest (2021–2022 bear market, 5 bps net friction).

1 • EVENT STUDY VALIDATION (|AR0| SHOCKS)



Mean impact scores per quintile shown in parentheses (N = 2,072; ~414 per bucket)

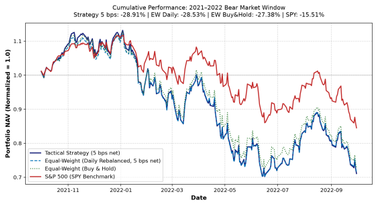
Statistical Evidence & Robustness

`ρ = 0.159`	Spearman rank correlation, 95% CI [0.122, 0.194], p = 3.08e-13.
`ρ = 0.116`	Partial correlation (p = 1.05e-07), controlling for trailing 20d volatility.
`p = 0.0010`	Placebo test: 1,000 within-ticker shuffles (null mean ρ = 0.028).

Finding: Impact score successfully ranks market surprise magnitude. However, correlation is not causation: |AR0| measures size of price move, not direction or tradeability.

2 • TACTICAL REBALANCING BACKTEST (MODULE A)

Cumulative normalized NAV (252 trading days, Oct 2021 – Sep 2022)



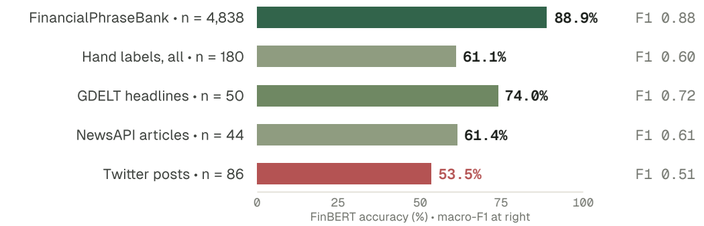
Strategy	Return	Sharpe	Max DD
Tactical (5 bps)	-28.91%	-0.9209	-37.95%
Equal-wt daily	-28.53%	-0.9092	-37.39%
Equal-wt buy & hold	-27.38%	-0.9323	-35.56%
SPY (S&P 500)	-15.51%	-0.7085	-24.37%

Honesty diagnostics:

- **200-run Placebo:** Sharpe sits at 48.5th percentile (near centre of random distribution).
- **TSLA exclusion:** -28.65% net return (dropping ~70% of tweets does not change result).
- **Sensitivity grid (9 cells):** Returns remain flat (-28.56% to -29.20%) across k ∈ [0.75, 3.0].

Where the model is weak, and what could come next

MODEL LIMITATION • MEASURED RESULT



A model that performs well on curated financial headlines may perform differently on informal social posts and other real-world text. Informal language, ambiguity and writing style are plausible causes, but they are hypotheses, not tested here.

FinBERT is precise on bad news (0.85), useful for risk detection, but misses subtle positive analyst targets (positive recall 0.30). Accuracy = share of texts labelled correctly; macro-F1 averages per-class scores.

PROPOSED EXTENSION • MODULE B, WHOLESALE LOAN STRESS TESTING

Proposed, not implemented or validated.

Risk signal • the same RiskSignal feed, zero engine modification

↓ **Credit-risk assessment** • severe Credit Event / Regulatory triggers would raise PD; macro severities would lower sector recovery (raise LGD)

↓ **Scenario analysis** • re-estimate exposures after sudden news shocks

↓ **Potential stress-testing application** • capital adequacy and regulatory stress reporting

$$EL = EAD \times PD_{adjusted} \times LGD_{adjusted}$$

Expected loss = how much is at risk × how likely default is × how much is lost if it happens.

PD Probability of Default • **LGD** Loss Given Default • **EAD** Exposure at Default

The project demonstrates an end-to-end path from financial text to explainable portfolio signals. Further validation is needed before treating those signals as a reliable source of investment advantage.